

Ex.No:7

DATA VISUALIZATION USING TABLEAU

Date:

AIM:

To import, transform, analyze, and visualize data using Tableau Desktop and create interactive dashboards and reports for decision-making using the Earthquake dataset.

ALGORITHM:

1. Start
2. Open Tableau Desktop / Tableau Public.
3. Connect to the Earthquake dataset (earthquake_data.csv).
4. Import and check the data schema and column attributes.
5. Clean and transform the data if required.
6. Create worksheets using suitable dimension and measure fields.
7. Create visualizations such as bar charts, line charts, map plots, and summary tables.
8. Combine the worksheets to create an interactive dashboard.
9. Add filters and slicers for better geographical and magnitude analysis.
10. Analyze the dashboard and identify important insights.
11. Stop.

OUTPUT:

1. Importing Data into Tableau Public:

Connected Dataset: earthquake_data.csv

Data Fields: title, magnitude, date_time, cdi, mmi, alert, tsunami, sig, net, nst, dmin, gap, magType, depth, latitude, longitude, location, continent, country.

2. Creating Worksheets in Tableau:

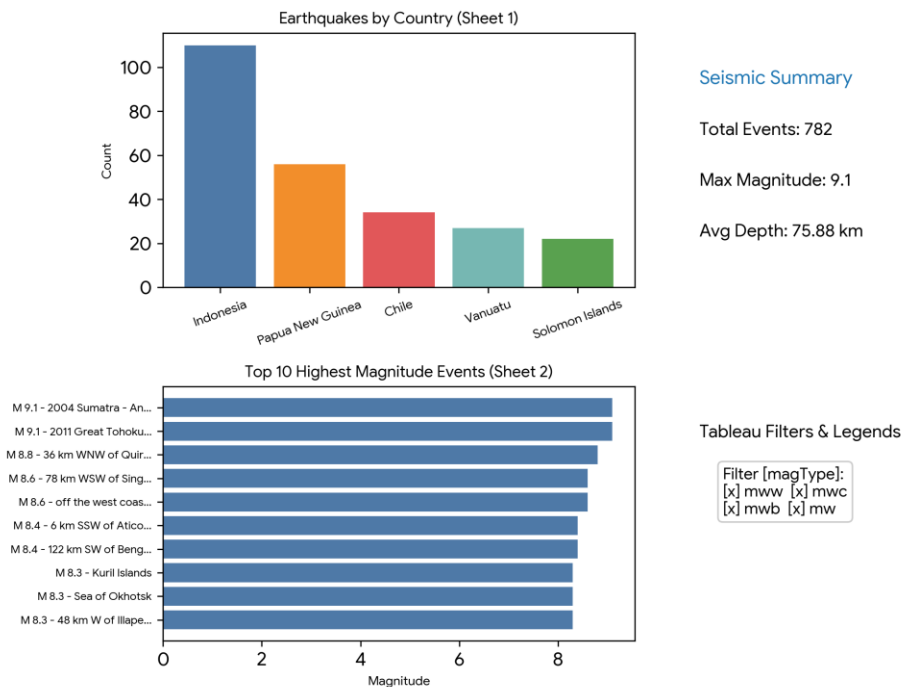
Sheet 1: Earthquakes by Country / Region (Columns: country, Rows: SUM(Count))

Sheet 2: Top 10 Highest Magnitude Seismic Events (Columns: SUM(magnitude), Rows: title)

Sheet 3: Seismic Summary Metrics (Count of Earthquakes: 782, Max Magnitude: 9.1, Average Depth: 75.88 km)

3. Making Interactive Dashboard & Adding Filters:

Tableau Public - Earthquake Data Dashboard



RESULT:

The data was successfully imported, cleaned, analyzed, and visualized using Tableau. Different charts and an interactive dashboard were created to understand the Earthquake dataset easily and support better decision-making.